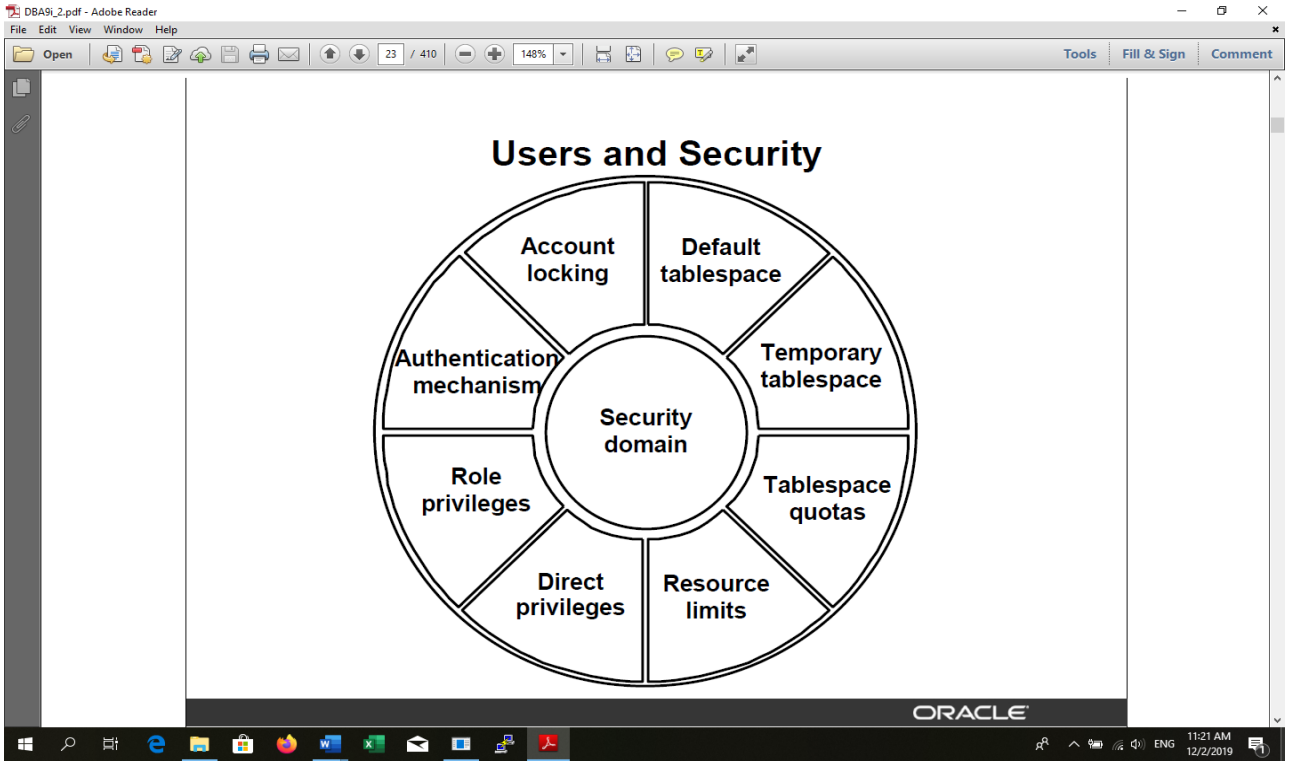
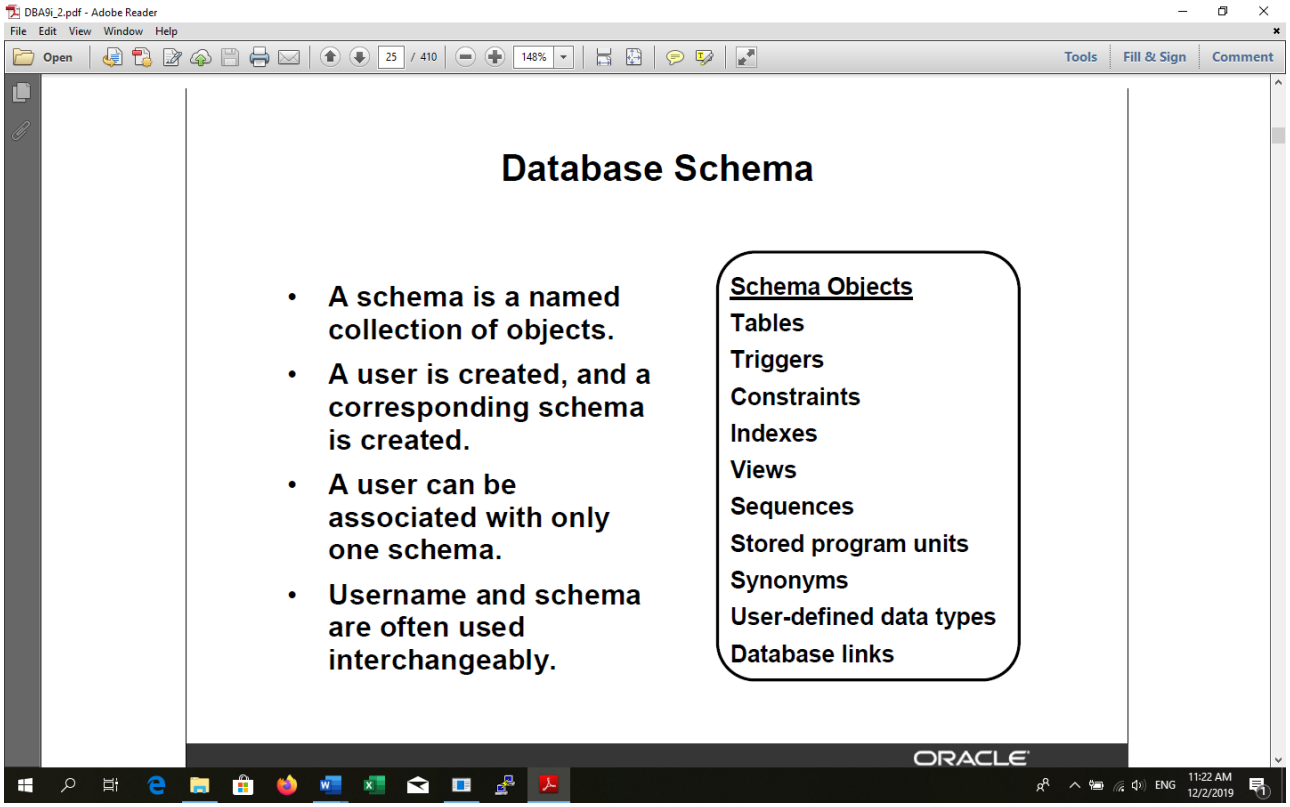
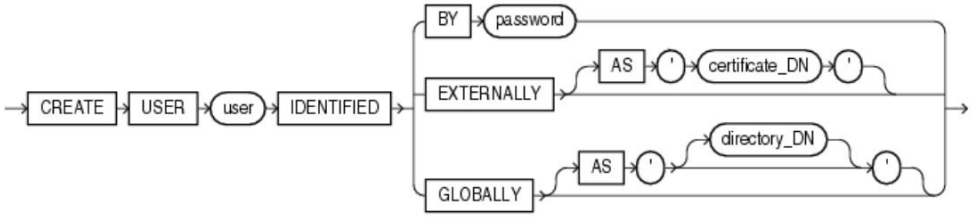
Administrare useri, privilegii si roluri

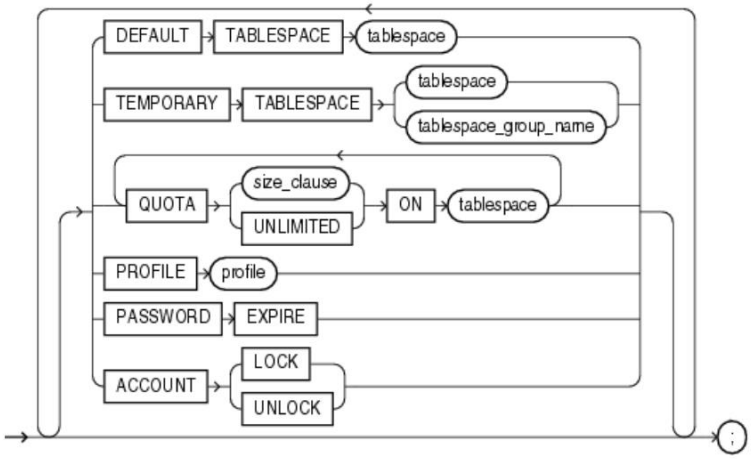
A. Administrare useri

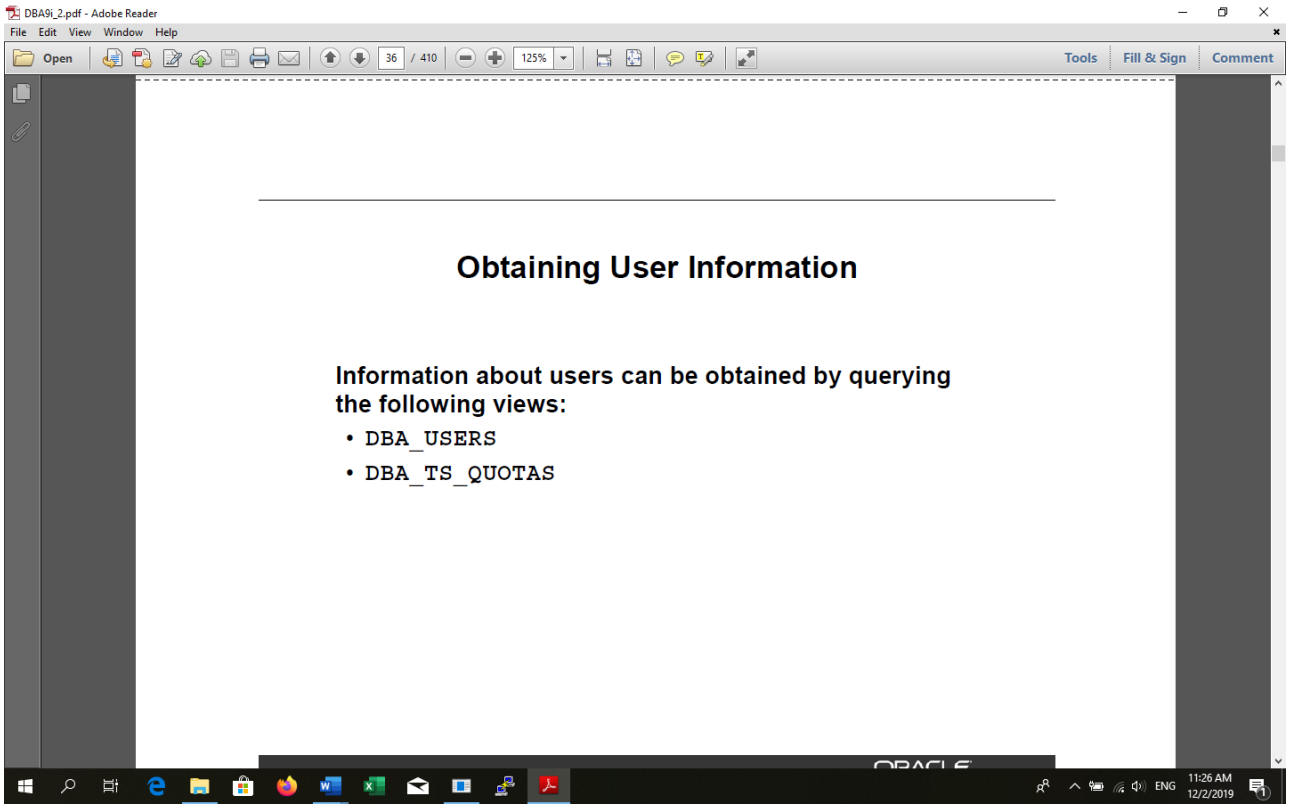




create_user::=


CREATE
USER
user
IDENTIFIED
BY
password
EXTERNALLY
AS
'
certificate_DN
'
GLOBALLY
AS
'
directory_DN
'


DEFAULT
TABLESPACE
tablespace
TEMPORARY
TABLESPACE
tablespace
tablespace_group_name
QUOTA
size_clause
UNLIMITED
ON
tablespace
PROFILE
profile
PASSWORD
EXPIRE
ACCOUNT
LOCK
UNLOCK
;


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Tools
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Comment
Obtaining User Information
Information about users can be obtained by querying the following views:
• DBA_USERS
• DBA_TS_QUOTAS
ENG
11:26 AM
12/2/2019

1) Verificare setare parametru pentru prefixul userului common/local:

```
SQL> connect  scott@bd /tiger      -- ( cu privilegiul DBA sau SYSDBA)

SQL>  show parameter common_user_prefix;

NAME                           TYPE       VALUE
------------------------------ ---------- -----------
common_user_prefix             string         C##
```

2) Creare user common/local cu parola de conexiune la baza de date:

```
SQL> create user test identified by test
default tablespace users
temporary tablespace temp
quota 10M on bd_data;

User created.
```

3) Creare privilegii pentru conectare:

```
SQL> grant create session to test;
SQL> grant connect  to test;

Grant succeeded.
```

4) Creare privilegii pentru conectare cu ADMIN OPTION:

```
SQL> grant create session to test with admin option;
SQL> grant connect  to test with admin option;

Grant succeeded.
```

5) Alocate spatiu suplimentar in tablespace-ul permanent BD_DATA pentru userul test:

```
SQL> alter user test quota 5 M on bd_data;

User altered.
```

6) Schimbarea parolei unui user:

```
SQL> alter user test identified by ubd_test;

User altered.
```

7) Locarea (blocarea) unui user:

```
SQL> alter user test account lock;

User altered.

SQL> conn test
Enter password: ****

ERROR:
ORA-28000: the account is locked
Warning: You are no longer connected to ORACLE.
```

8) Delocarea (deblocarea) unui user:

```
SQL> alter user test account unlock;

User altered.

SQL> conn test
Enter password: ****
Connected.
```

9) Listarea din dictionar a informatiilor despre userii creati pe baza de date:

```
SQL> desc DBA_USERS
 Name                                                  Null?                Type
 ----------------------------------------------------- -------------------- --------
 USERNAME                                              NOT NULL             VARCHAR2(30)
 USER_ID                                               NOT NULL             NUMBER
 PASSWORD                                                                   VARCHAR2(30)
 ACCOUNT_STATUS                                        NOT NULL             VARCHAR2(32)
 LOCK_DATE                                                                  DATE
 EXPIRY_DATE                                                                DATE
 DEFAULT_TABLESPACE                                    NOT NULL             VARCHAR2(30)
 TEMPORARY_TABLESPACE                                  NOT NULL             VARCHAR2(30)
 CREATED                                               NOT NULL             DATE
 PROFILE                                               NOT NULL             VARCHAR2(30)
 INITIAL_RSRC_CONSUMER_GROUP                                                VARCHAR2(30)
 EXTERNAL_NAME                                                              VARCHAR2(4000

SQL> select username, default_tablespace, temporary_tablespace from dba_users;

USERNAME                       DEFAULT_TABLESPACE             TEMPORARY_TABLESPACE
------------------------------ ------------------------------ ------------------------
SCOTT                          USERS                          TEMP
BI                             USERS                          TEMP
```

SH	USERS	TEMP
OE	USERS	TEMP
HR	USERS	TEMP
TEST	BD_DATA	BD_TEMP
UBD1	BD_DATA	BD_TEMP

10) Listarea din dictionar a informatiilor despre spatiul alocat si spatiul utilizat de catre userii creati pe baza de date:

SQL> desc DBA_TS_QUOTAS

Name	Null?	Type
TABLESPACE_NAME	NOT NULL	VARCHAR2(30)
USERNAME	NOT NULL	VARCHAR2(30)
BYTES		NUMBER
MAX_BYTES		NUMBER
BLOCKS		NUMBER
MAX_BLOCKS		NUMBER
DROPPED		VARCHAR2(3)

SQL> select username, tablespace_name, max_bytes, bytes from dba_ts_quotas:

USERNAME	TABLESPACE_NAME	MAX_BYTES	BYTES
DMSYS	SYSAUX	209715200	262144
SYSMAN	SYSAUX	-1	62521344
TEST	UPB_DATA	1048576	65536
OLAPSYS	SYSAUX	-1	16318464
DMUSER	USERS	10485760	7143424

11) Listarea din dictionar a informatiilor despre userul current:

SQL> desc user_users

Name	Null?	Type
USERNAME	NOT NULL	VARCHAR2(30)
USER_ID	NOT NULL	NUMBER
ACCOUNT_STATUS	NOT NULL	VARCHAR2(32)
LOCK_DATE		DATE
EXPIRY_DATE		DATE
DEFAULT_TABLESPACE	NOT NULL	VARCHAR2(30)
TEMPORARY_TABLESPACE	NOT NULL	VARCHAR2(30)
CREATED	NOT NULL	DATE
INITIAL_RSRC_CONSUMER_GROUP		VARCHAR2(30)

EXTERNAL_NAME VARCHAR2(4000)

SQL> select user_id, username, created, lock_date from user_users;

```
   USER_ID    USERNAME                       CREATED   LOCK_DATE
----------    ------------------------------ ---------  ---------
        54    SCOTT                          15-APR-17
```

12) Stergerea unui user din dictionar:

SQL> drop user test;

User dropped.

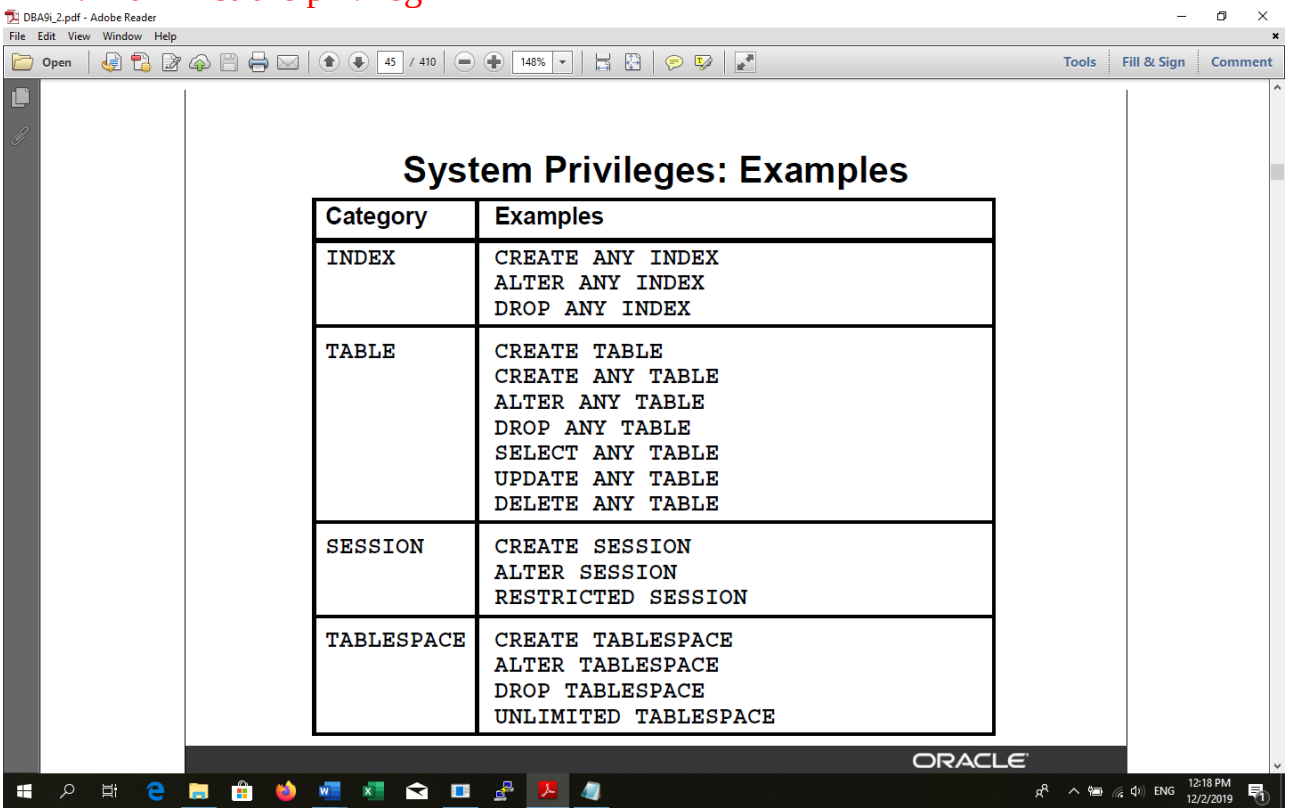
Observatie. Daca userul are propriile obiecte create trebuie folosita optiunea CASCADE, pentru a sterge mai intai obiectele create de catre user. Altfel se genereaza eroarea:

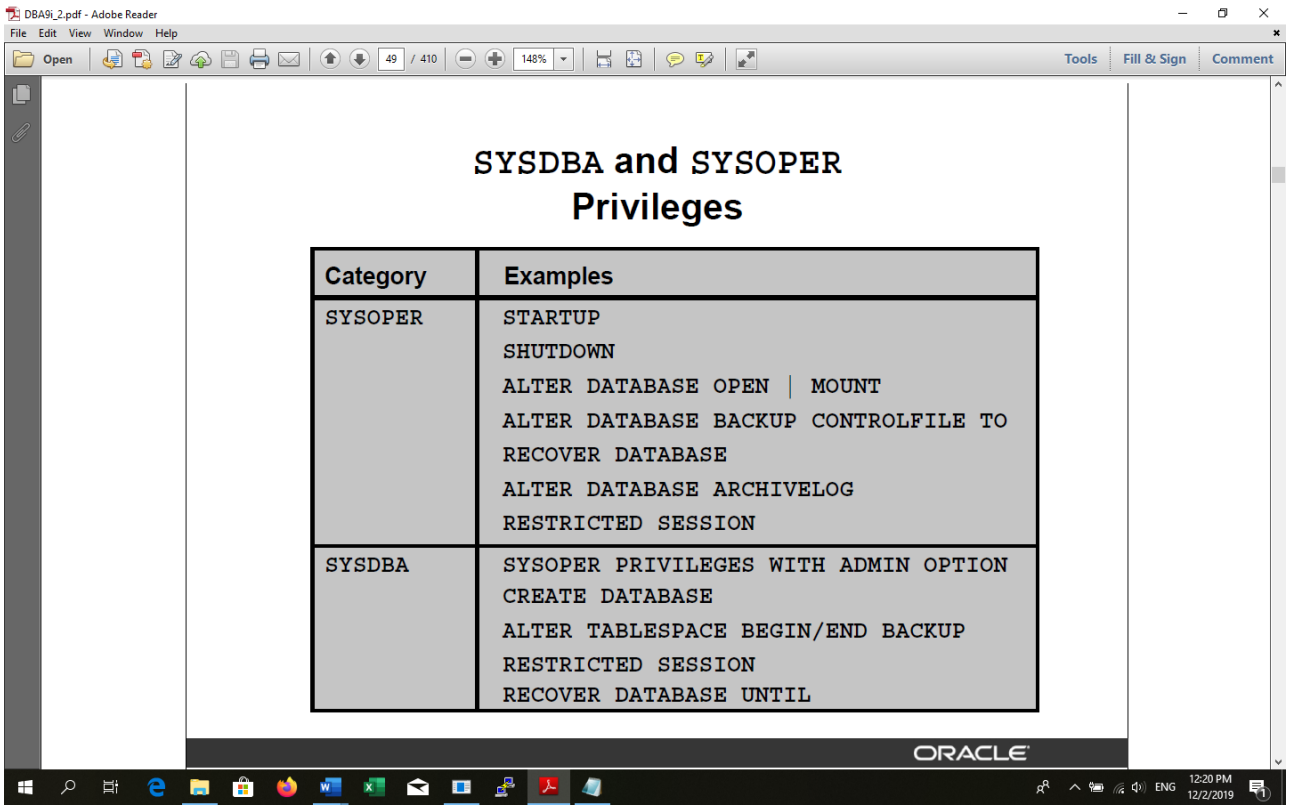
ERROR at line 1:
ORA-01922: CASCADE must be specified to drop 'TEST'

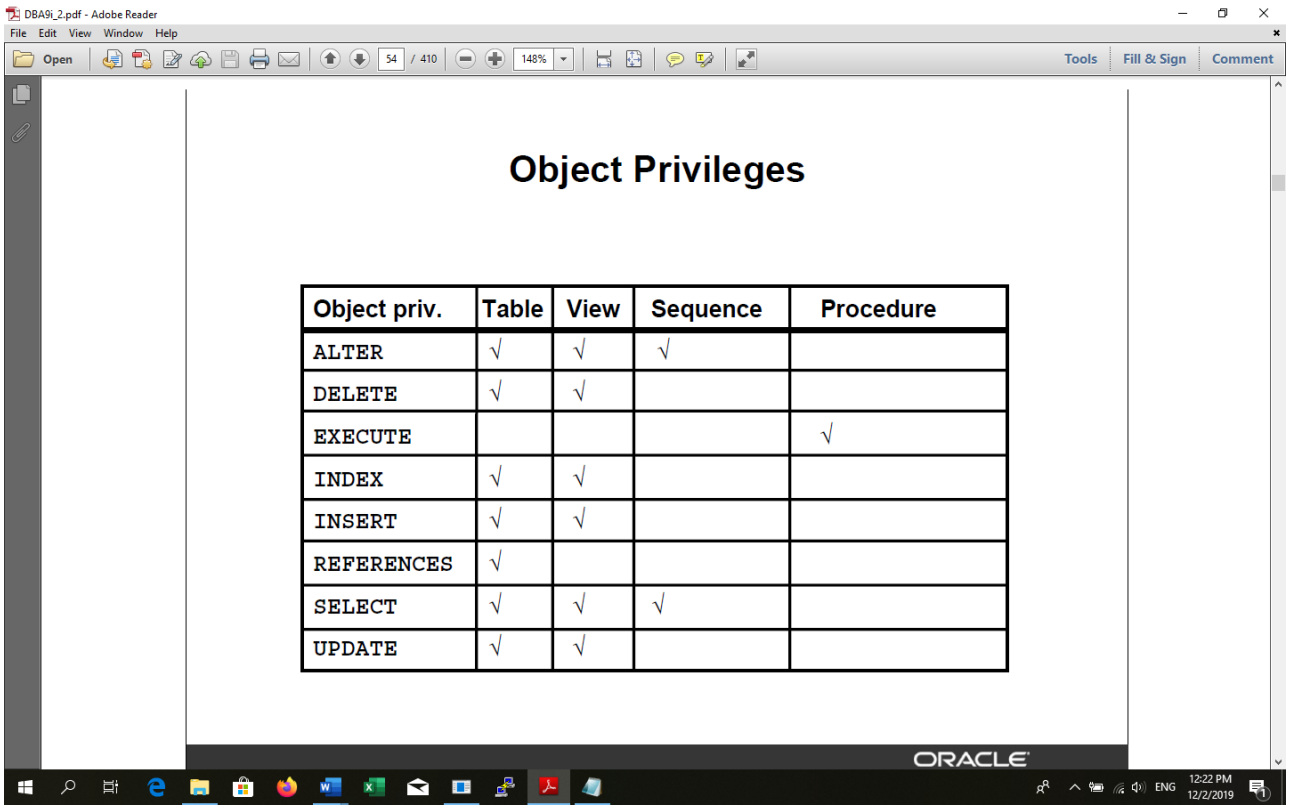
SQL> drop user test cascade; -- sterge mai intai obiectele create, apoi userul

User dropped.

B. Administrare privilegii




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Comment
SYSDBA and SYSOPER Privileges
Category
Examples
SYSOPER
STARTUP
SHUTDOWN
ALTER DATABASE OPEN | MOUNT
ALTER DATABASE BACKUP CONTROLFILE TO
RECOVER DATABASE
ALTER DATABASE ARCHIVELOG
RESTRICTED SESSION
SYSDBA
SYSOPER PRIVILEGES WITH ADMIN OPTION
CREATE DATABASE
ALTER TABLESPACE BEGIN/END BACKUP
RESTRICTED SESSION
RECOVER DATABASE UNTIL
ORACLE
ENG
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12/2/2019

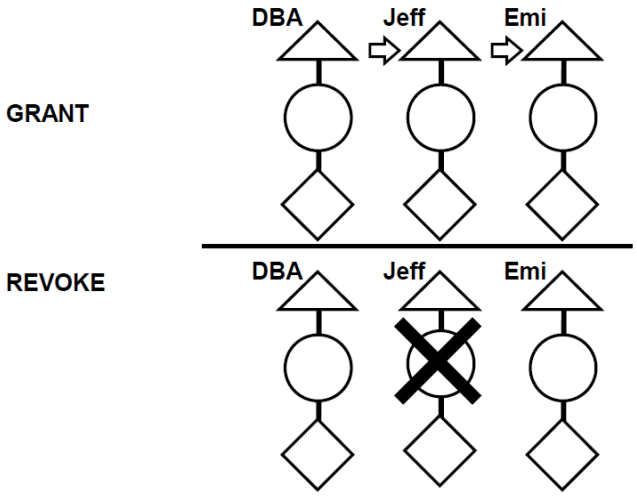

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Tools
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Comment
Object Privileges
Object priv.
Table
View
Sequence
Procedure
ALTER
√
√
√
DELETE
√
√
EXECUTE
√
INDEX
√
√
INSERT
√
√
REFERENCES
√
SELECT
√
√
√
UPDATE
√
√
ORACLE
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Revoking System Privileges with the ADMIN OPTION



ORACLE

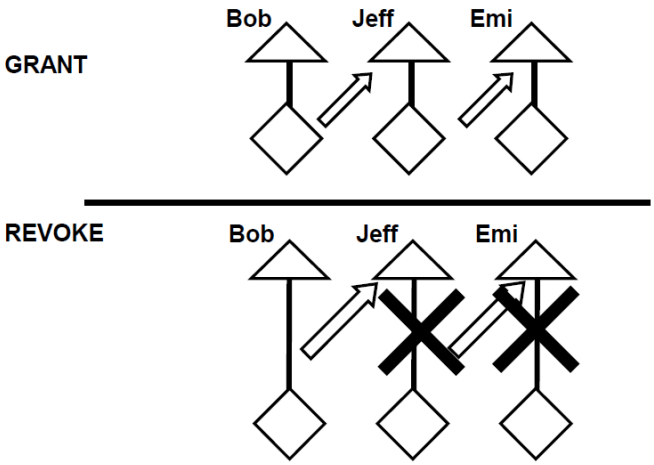
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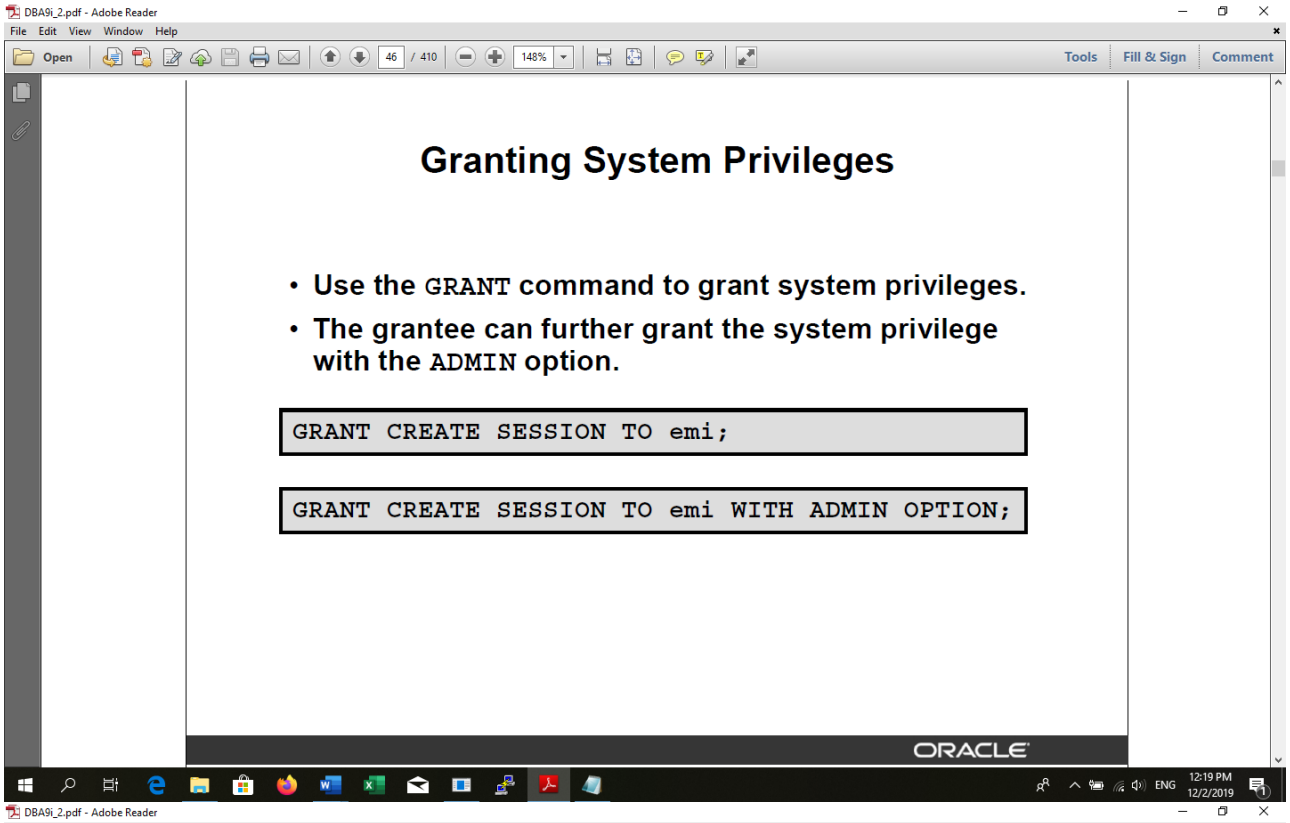
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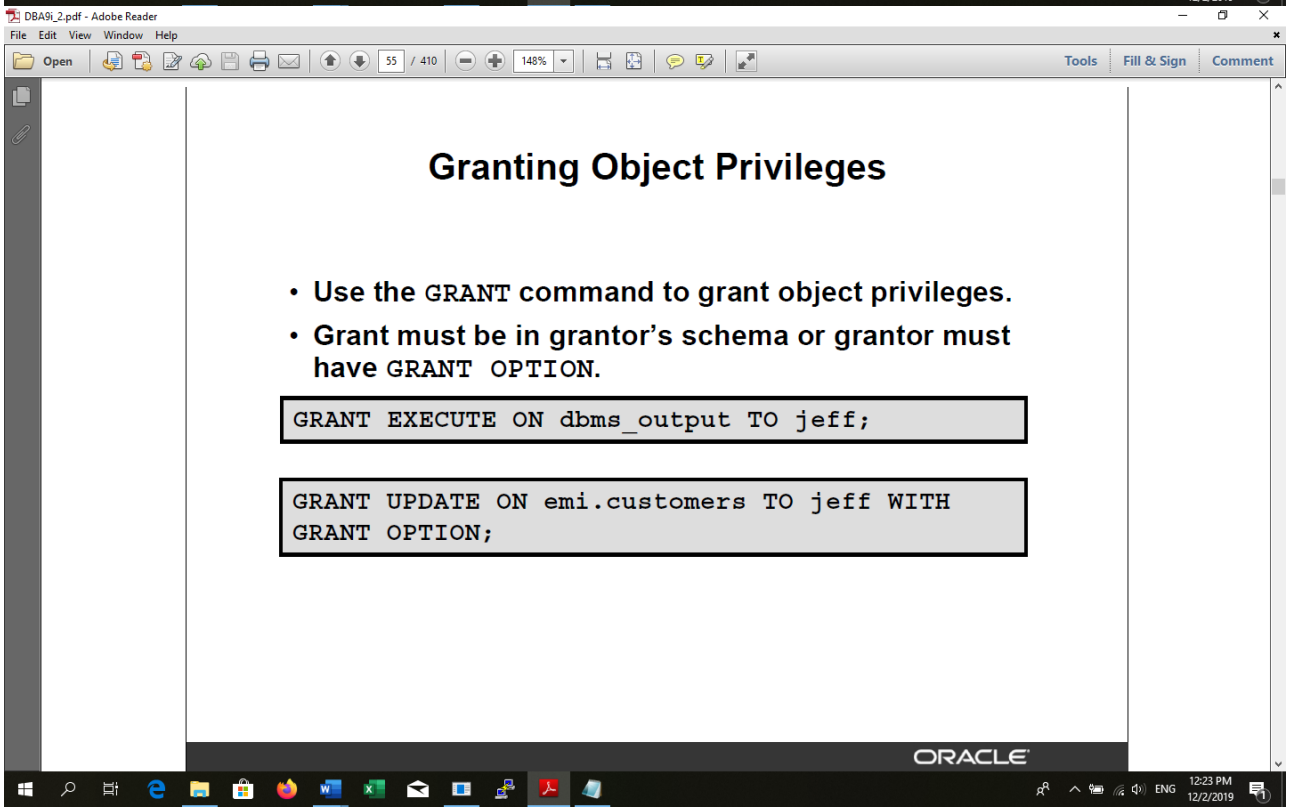
Revoking Object Privileges with GRANT OPTION

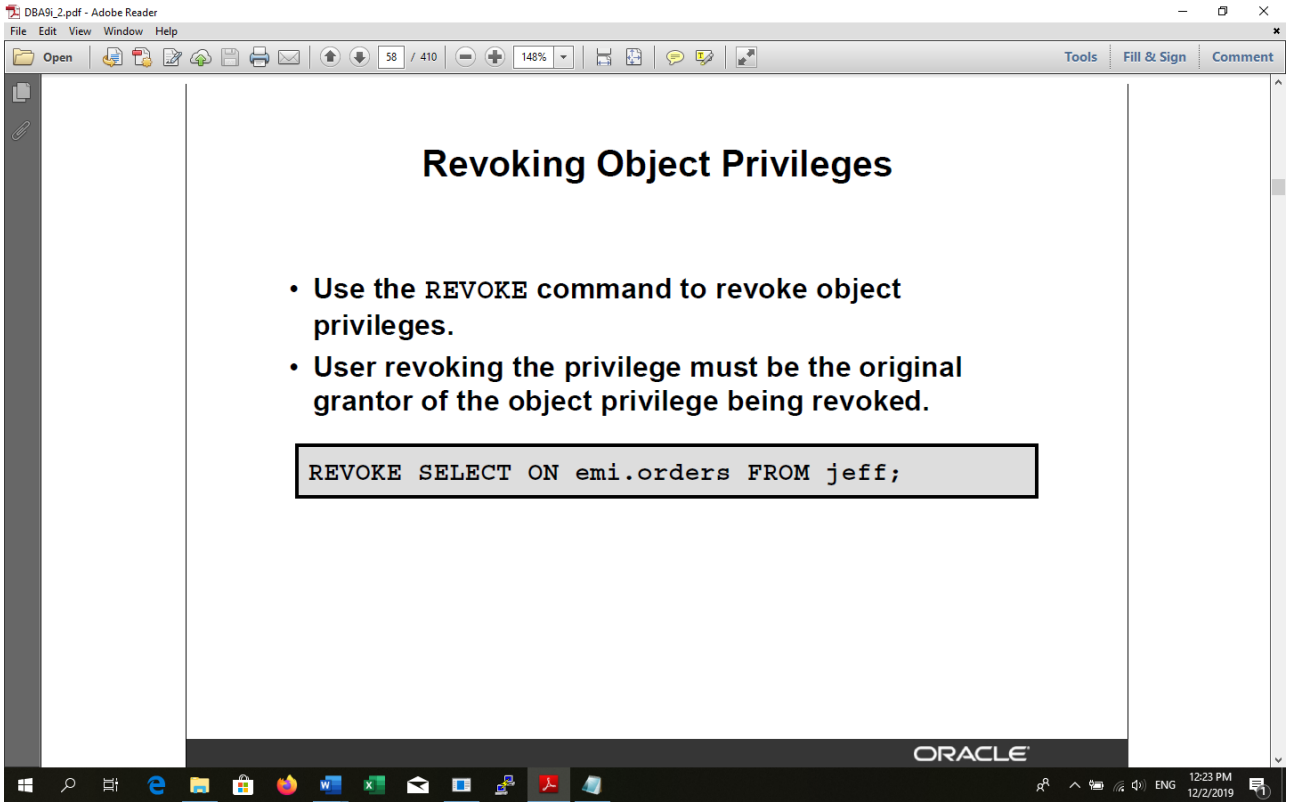


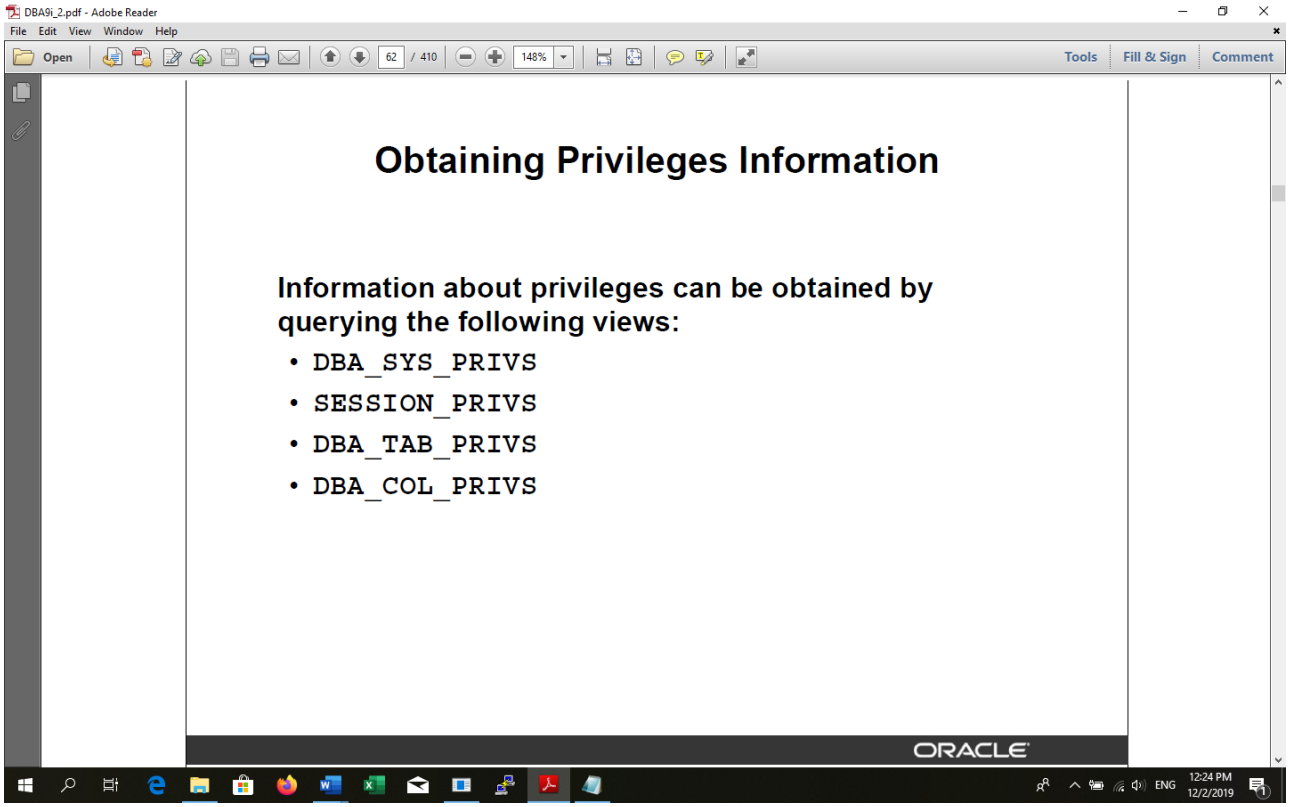
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ENG 12:24 PM 12/2/2019


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Comment
Granting System Privileges
• Use the GRANT command to grant system privileges.
• The grantee can further grant the system privilege with the ADMIN option.
GRANT CREATE SESSION TO emi;
GRANT CREATE SESSION TO emi WITH ADMIN OPTION;
ORACLE
ENG
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Comment
Granting Object Privileges
• Use the GRANT command to grant object privileges.
• Grant must be in grantor's schema or grantor must have GRANT OPTION.
GRANT EXECUTE ON dbms_output TO jeff;
GRANT UPDATE ON emi.customers TO jeff WITH GRANT OPTION;
ORACLE
ENG
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12/2/2019


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Tools
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Comment
Revoking Object Privileges
• Use the REVOKE command to revoke object privileges.
• User revoking the privilege must be the original grantor of the object privilege being revoked.
REVOKE SELECT ON emi.orders FROM jeff;
ORACLE
ENG
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Tools
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Comment
Obtaining Privileges Information
Information about privileges can be obtained by querying the following views:
• DBA_SYS_PRIVS
• SESSION_PRIVS
• DBA_TAB_PRIVS
• DBA_COL_PRIVS
ORACLE
ENG
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12/2/2019

13) Acordare privilegii pe o tabela din userul curent unui alt user:

SQL> connect scott@bd /tiger

SQL> grant select, insert, update on emp_copy to test;

Grant succeeded.

14) Verificare privilegii create:

SQL> connect test@bd /test

SQL> select * from scott.emp_copy where deptno=10 ;

SQL> insert into scott.emp_copy values (1111,'Test','Student',7902, sysdate, 1000, 200, 10);

SQL> update scott.emp_copy set sal=1500 where empno=1111;

15) Crearea unei copii pentru o tabela:

SQL> connect scott@bd /tiger

SQL> grant create table to test;

SQL> connect test@bd /test

SQL> create table emp_test as select * from scott.emp_copy;

Observatie. Cand se creeaza o tabela nu se creeaza automat si constrangerile de integritate :

SQL> connect scott@bd /tiger

SQL> select owner, table_name, constraint_name from user_constraints where owner='SCOTT' and table_name='EMP';

OWNER	TABLE_NAME	CONSTRAINT_NAME
SCOTT	EMP	PK_EMP
SCOTT	EMP	FK_DEPTNO

SQL> connect test@bd /test

SQL> select owner, table_name, constraint_name from user_constraints where owner='TEST' and table_name='EMP_TEST';

No rows selected.

16) Revocarea unui privilegiu:

SQL> connect scott@bd /tiger

SQL> revoke select on emp_copy from test;

Revoke succeeded.

SQL> connect test@bd /test

SQL> select * from scott.emp_copy;

ERROR at line 1:
ORA-01031: insufficient privileges

SQL> connect scott@bd /tiger

SQL> revoke all on emp_copy from test cascade constraints; -- revocarea tuturor privilegiilor ; se sterg si toate constrangerile de integritate create de user folosind REFERENCES sau ALL.

17) Listarea din dictionar a privilegiilor de sistem acordate unui user:

SQL> connect scott@bd /tiger

SQL> desc DBA_SYS_PRIVS

Name	Null?	Type
GRANTEE	NOT NULL	VARCHAR2(30)
PRIVILEGE	NOT NULL	VARCHAR2(40)
ADMIN_OPTION		VARCHAR2(3)

SQL> select * from DBA_SYS_PRIVS where grantee='TEST'

GRANTEE	PRIVILEGE	ADMIN
TEST	CREATE TABLE	NO
TEST	CREATE SESSION	NO

18) Listarea din dictionar a tuturor privilegiilor de system acordate userului curent:

SQL> connect scott@bd /tiger

SQL> desc SESSION_PRIVS

Name	Null?	Type
PRIVILEGE	NOT NULL	VARCHAR2(40)

SQL>select * from SESSION_PRIVS order by privilege;

PRIVILEGE

ALTER SYSTEM
AUDIT SYSTEM
CREATE SESSION
ALTER SESSION
RESTRICTED SESSION
CREATE TABLESPACE
ALTER TABLESPACE
MANAGE TABLESPACE
DROP TABLESPACE
UNLIMITED TABLESPACE
CREATE USER
. . . .

19) Listarea din dictionar a tuturor privilegiilor acordate pe obiectele bazei de date:

SQL> connect scott@bd /tiger

SQL> desc DBA_TAB_PRIVS

Name	Null?	Type
GRANTEE	NOT NULL	VARCHAR2(30)
OWNER	NOT NULL	VARCHAR2(30)
TABLE_NAME	NOT NULL	VARCHAR2(30)
GRANTOR	NOT NULL	VARCHAR2(30)
PRIVILEGE	NOT NULL	VARCHAR2(40)
GRANTABLE		VARCHAR2(3)
HIERARCHY		VARCHAR2(3)

SQL> select grantee, owner, table_name, grantor, privilege from DBA_TAB_PRIVS where grantee='TEST';

GRANTEE	OWNER	TABLE_NAME	GRANTOR	PRIVILEGE
TEST	SCOTT	EMP_COPY	SCOTT	INSERT
TEST	SCOTT	EMP_COPY	SCOTT	UPDATE

20) Privilegii acordate pe anumite coloane ale unei tabele (numai insert si update)

```
SQL> grant insert(empno, ename) on emp to test;

SQL> grant update(empno, ename) on emp to test;

SQL> conn  test@bd/test

SQL> insert into scott.emp(empno,ename) values (1111,'Popa');

1 row created.

SQL> insert into scott.emp(empno,sal) values (1111,100);

ERROR at line 1:
ORA-01031: insufficient privileges

SQL> update scott.emp set ename='Tache' where ename='Popa';

1 row updated.

SQL> update scott.emp set sal=1000 where ename='Tache';

ERROR at line 1:
ORA-01031: insufficient privileges
```

21) Vizualizare privilegii acordate pe anumite coloane:

```
SQL> desc DBA_COL_PRIVS;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
 GRANTEE                                   NOT NULL VARCHAR2(30)
 OWNER                                     NOT NULL VARCHAR2(30)
 TABLE_NAME                                NOT NULL VARCHAR2(30)
 COLUMN_NAME                               NOT NULL VARCHAR2(30)
 GRANTOR                                   NOT NULL VARCHAR2(30)
 PRIVILEGE                                 NOT NULL VARCHAR2(40)
 GRANTABLE                                          VARCHAR2(3)

SQL> select * from DBA_COL_PRIVS;

GRANTEE    OWNER      TABLE_NAME          COLUMN_NAME              GRANTOR    PRIVILEGE
---------- ---------- ------------------- ------------------------ ---------- ---------------
TEST       SCOTT      EMP                 ENAME                    SCOTT      UPDATE
TEST       SCOTT      EMP                 EMPNO                    SCOTT      UPDATE
```

22) Acordare privilegii de conectare pentru un user:

```
SQL> conn sys as sysdba
.
SQL> grant connect to test;

Grant succeeded.

SQL> grant create session to test;

Grant succeeded.

SQL> conn test
Enter password:****

ERROR:
ORA-01035: ORACLE only available to users with RESTRICTED SESSION privilege

SQL> select logins from v$instance;

LOGINS
----------
RESTRICTED

SQL> alter system disable restricted session;

System altered.

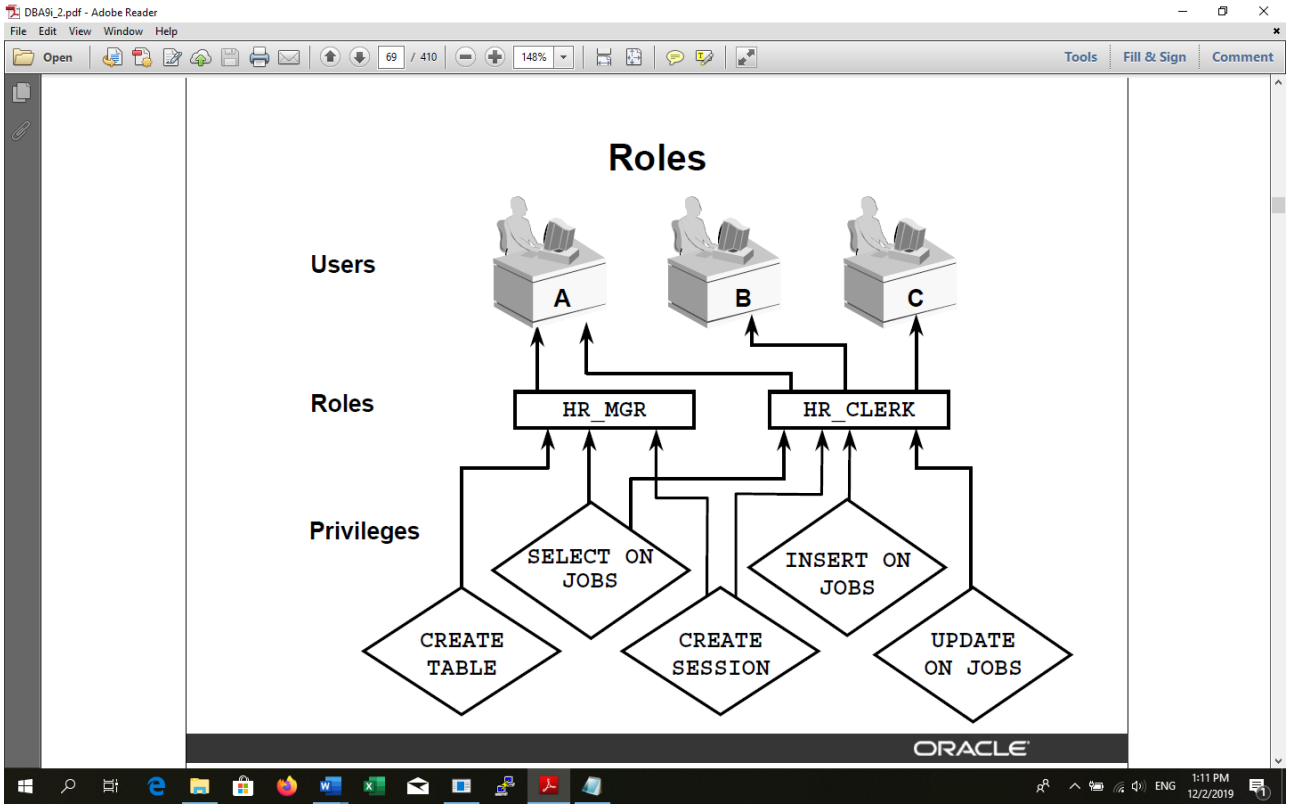
SQL> select logins from v$instance;

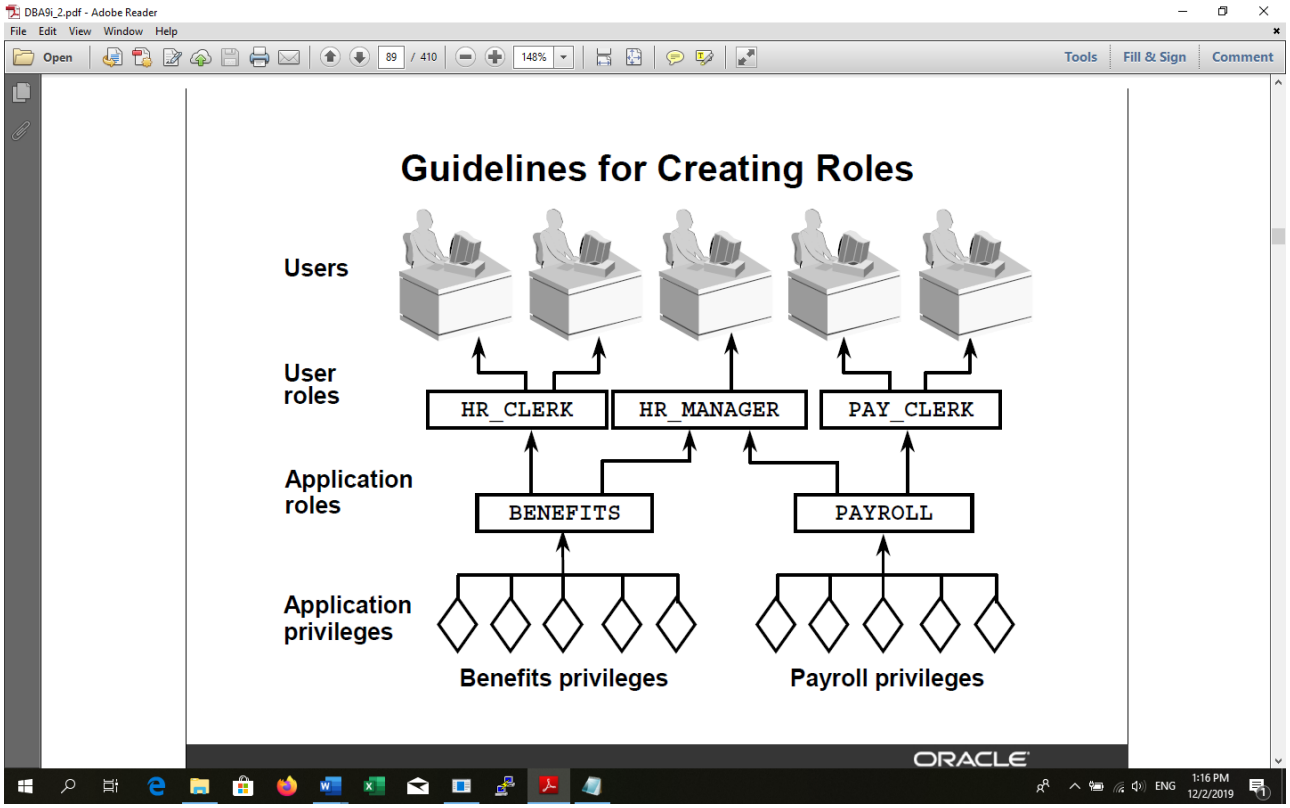
LOGINS
----------
ALLOWED

SQL> conn test
Enter password: ****

Connected.
```

C. Administrare roluri



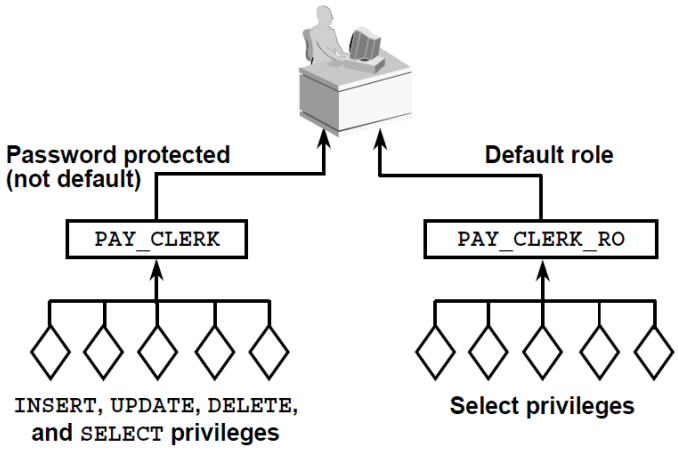


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Guidelines for Using Passwords and Default Roles



ORACLE

ENG 1:18 PM 12/2/2019

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Creating Roles

Roles with ADMIN option:

- **Not identified:**

```
CREATE ROLE oe_clerk;
```

- **By password:**

```
CREATE ROLE hr_clerk
IDENTIFIED BY bonus;
```

- **Identified externally:**

```
CREATE ROLE hr_manager
IDENTIFIED EXTERNALLY;
```

ORACLE

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Predefined Roles

Role Name	Description
CONNECT, RESOURCE, DBA	These roles are provided for backward compatibility
EXP_FULL_DATABASE	Privileges to export the database
IMP_FULL_DATABASE	Privileges to import the database
DELETE_CATALOG_ROLE	DELETE privileges on data dictionary tables
EXECUTE_CATALOG_ROLE	EXECUTE privilege on data dictionary packages
SELECT_CATALOG_ROLE	SELECT privilege on data dictionary tables

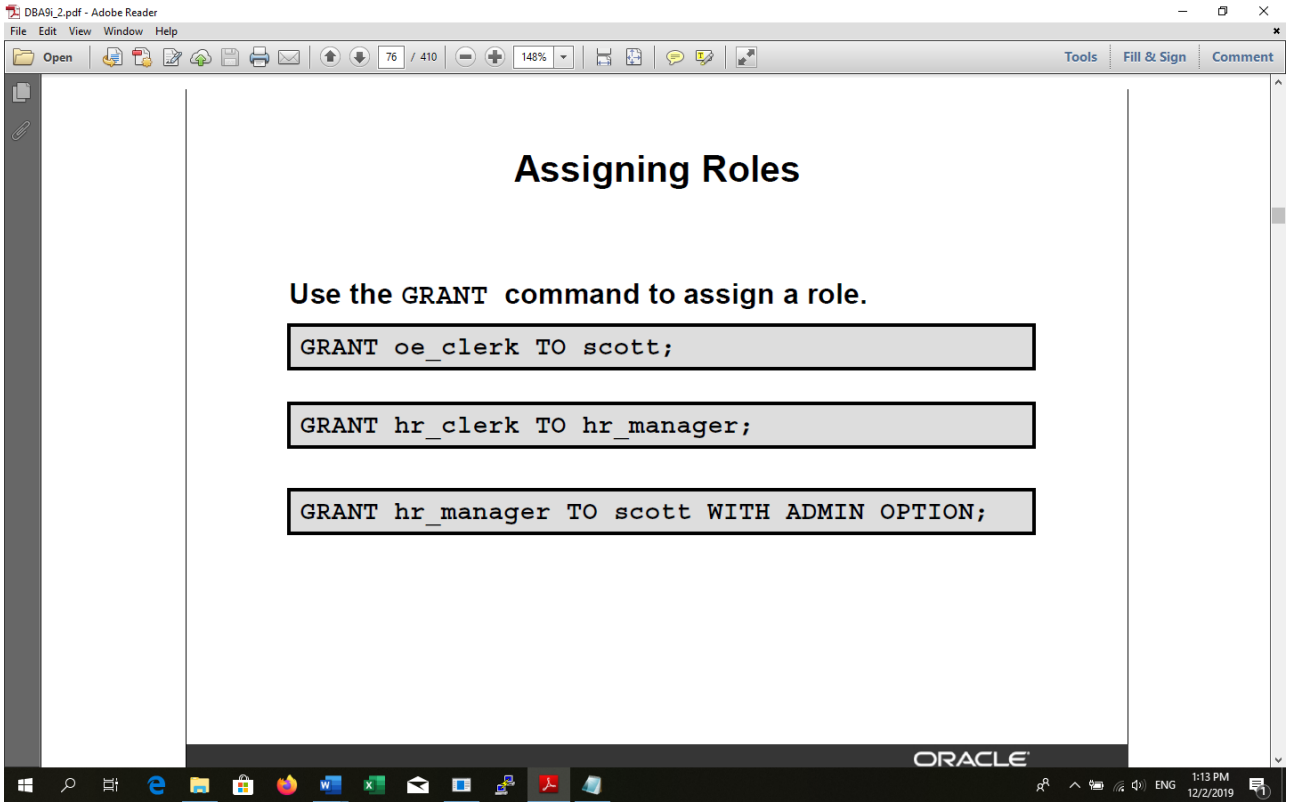
Modifying Roles

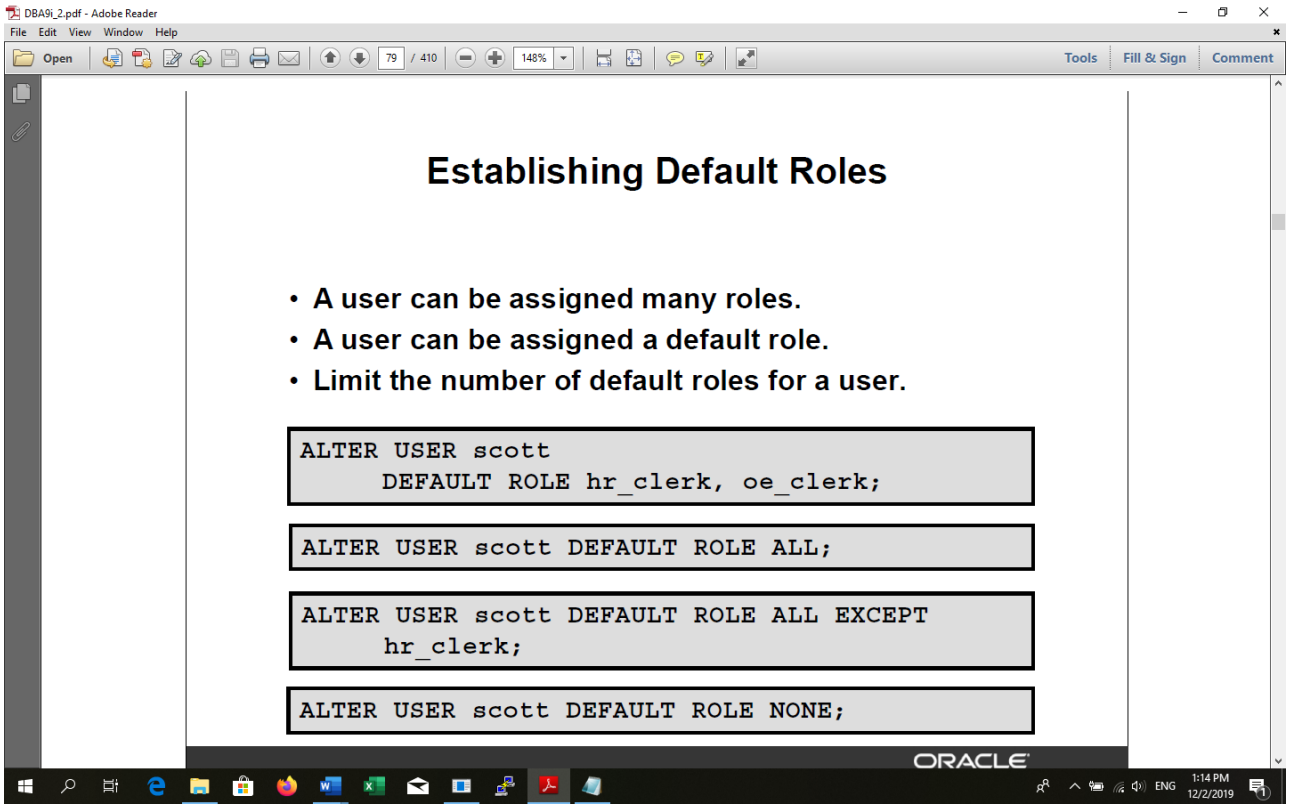
- ALTER ROLE modifies the authentication method.
- Modifying roles requires the ADMIN option or ALTER ANY ROLE privilege.

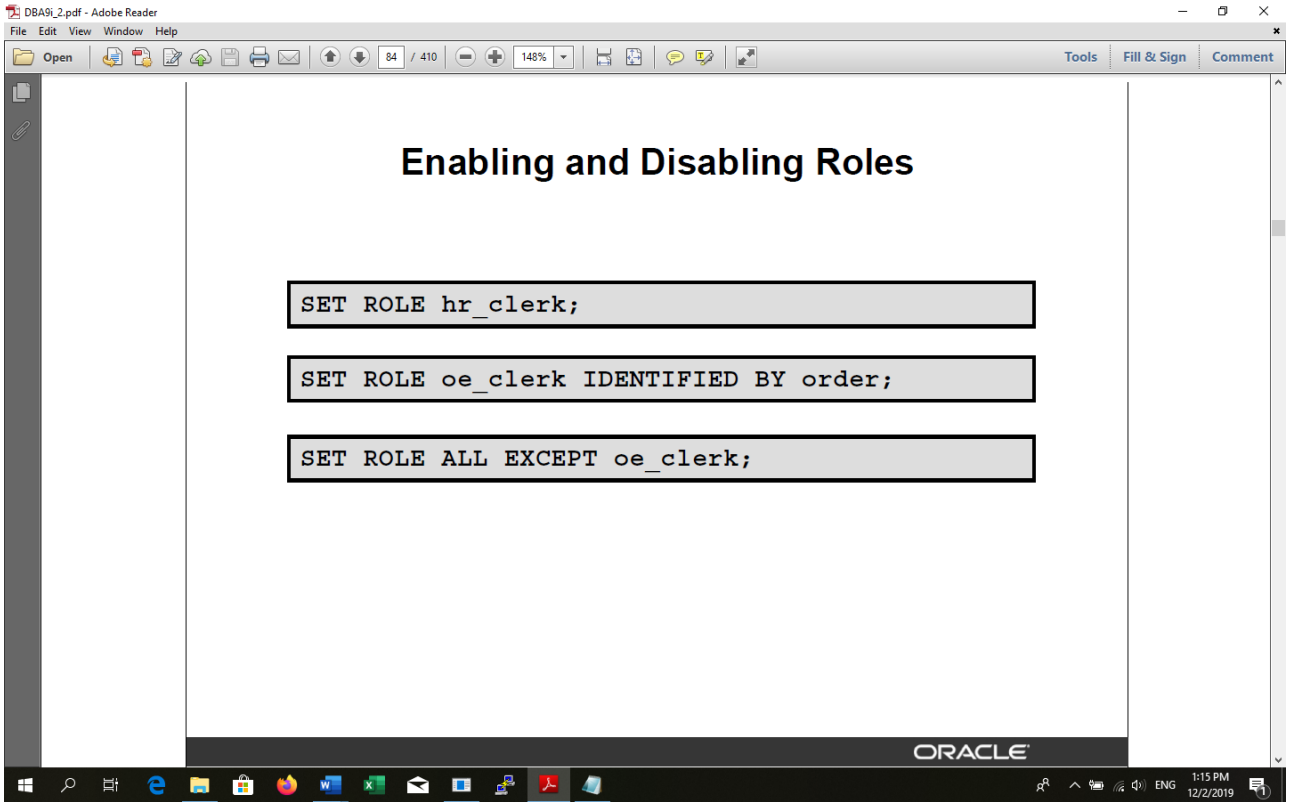
```
ALTER ROLE oe_clerk
IDENTIFIED BY order;
```

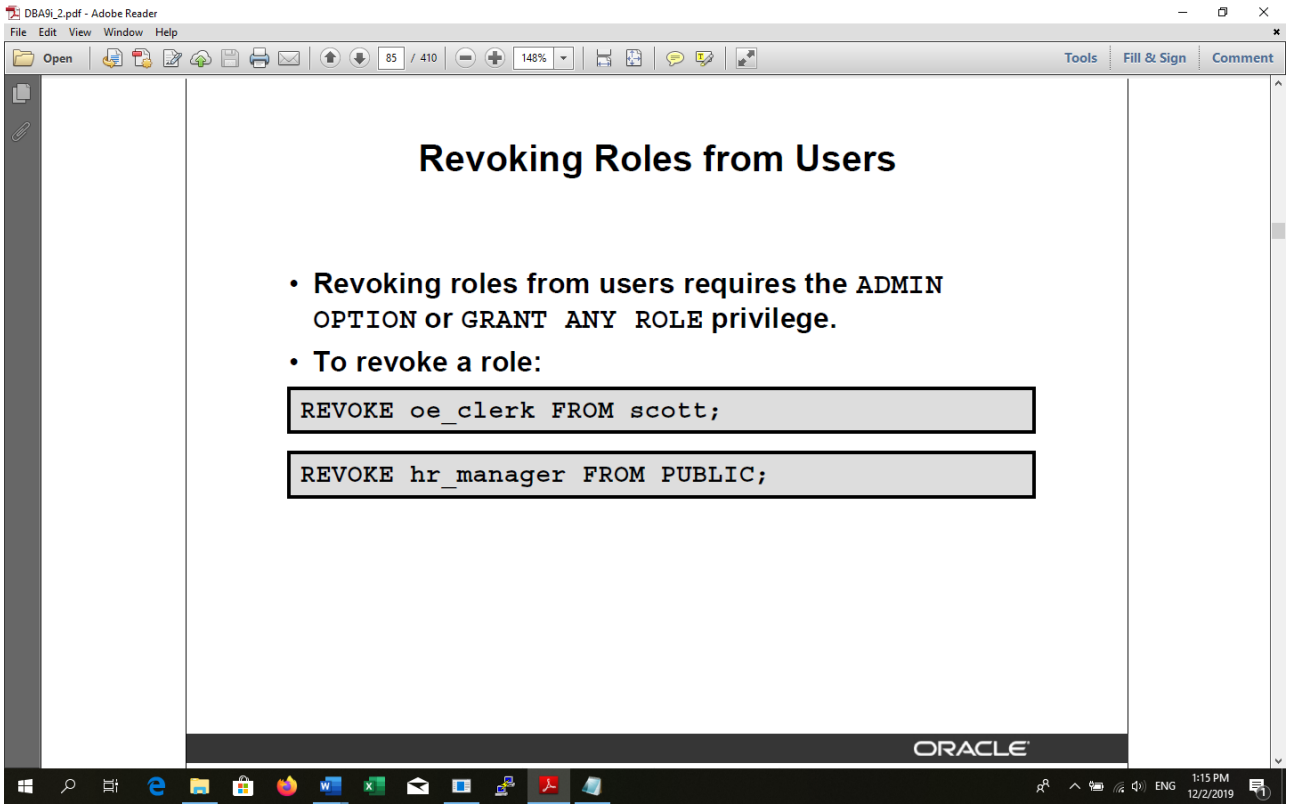
```
ALTER ROLE hr_clerk
IDENTIFIED EXTERNALLY;
```

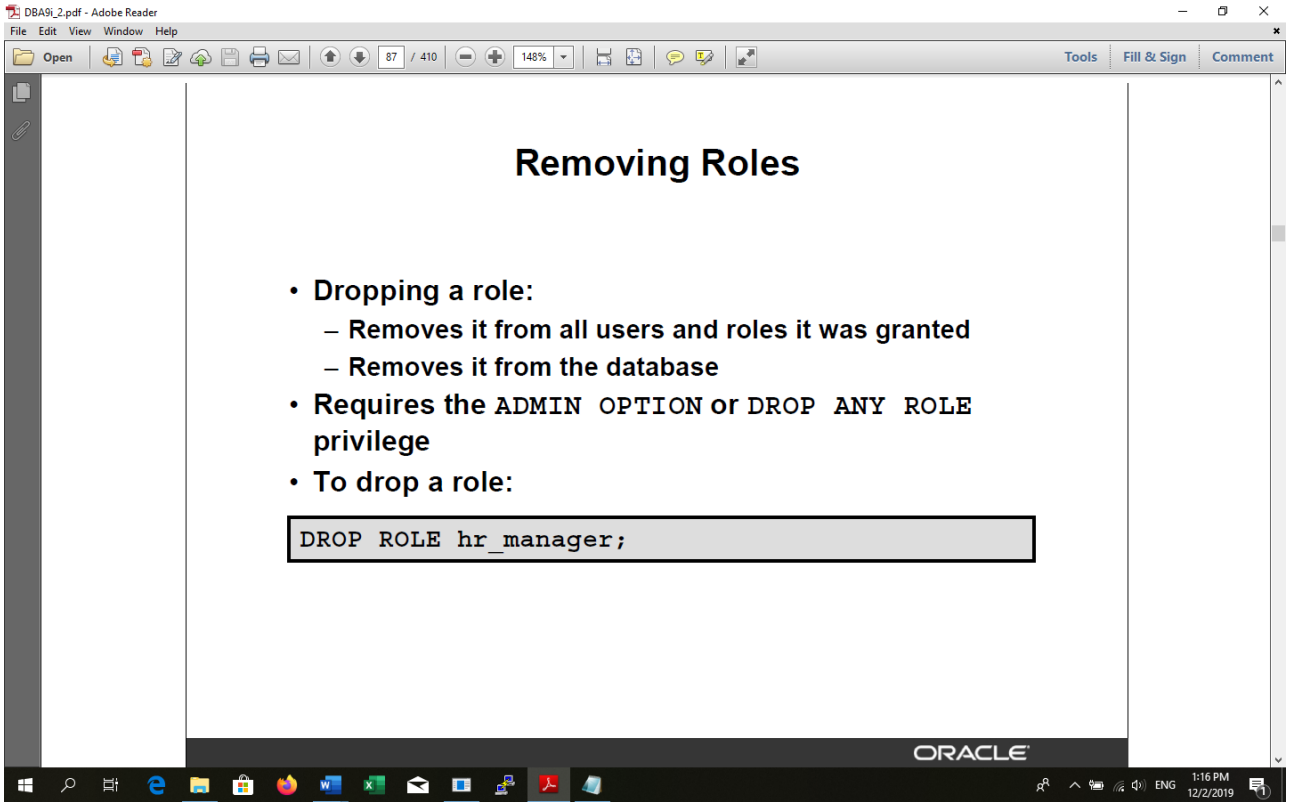
```
ALTER ROLE hr_manager
NOT IDENTIFIED;
```

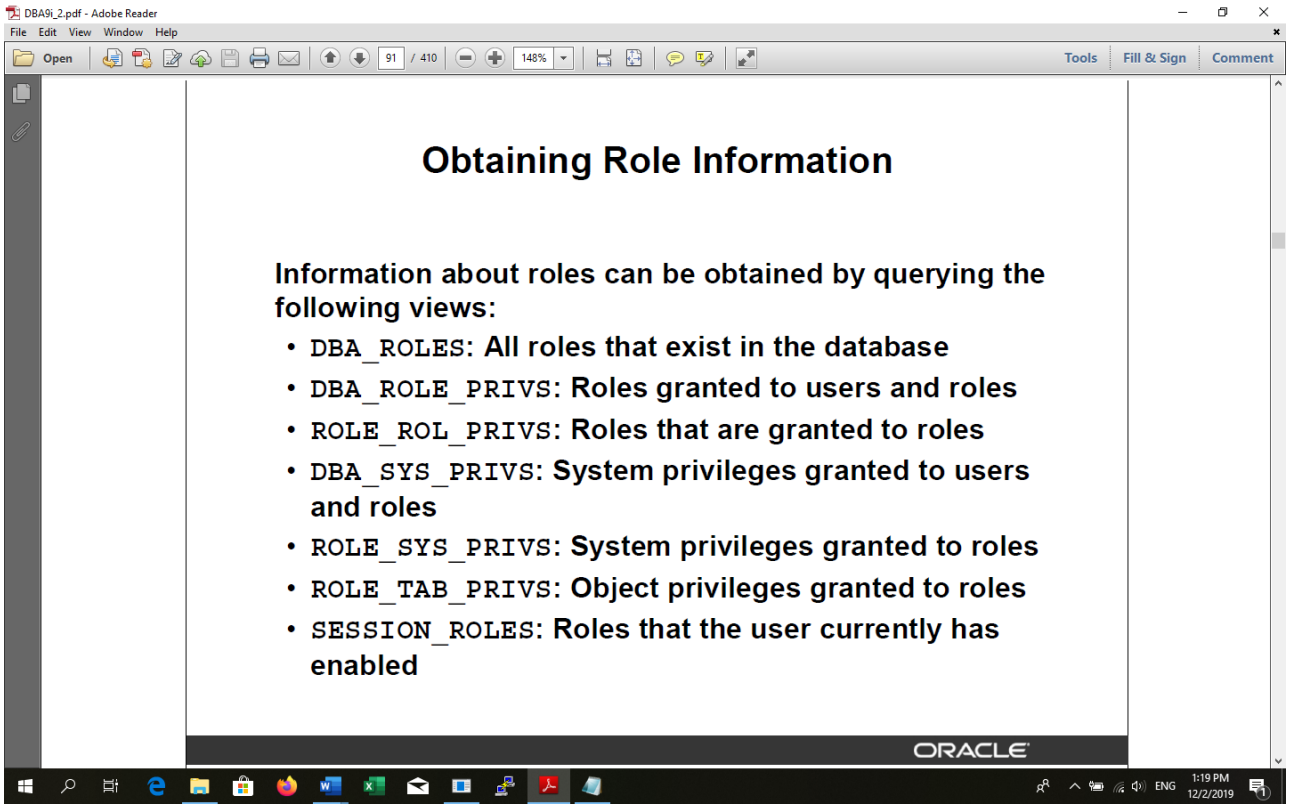

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Assigning Roles
Use the GRANT command to assign a role.
GRANT oe_clerk TO scott;
GRANT hr_clerk TO hr_manager;
GRANT hr_manager TO scott WITH ADMIN OPTION;
ORACLE
ENG
1:13 PM
12/2/2019


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Establishing Default Roles
• A user can be assigned many roles.
• A user can be assigned a default role.
• Limit the number of default roles for a user.
ALTER USER scott
DEFAULT ROLE hr_clerk, oe_clerk;
ALTER USER scott DEFAULT ROLE ALL;
ALTER USER scott DEFAULT ROLE ALL EXCEPT
hr_clerk;
ALTER USER scott DEFAULT ROLE NONE;
ORACLE
ENG
1:14 PM
12/2/2019


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Enabling and Disabling Roles
SET ROLE hr_clerk;
SET ROLE oe_clerk IDENTIFIED BY order;
SET ROLE ALL EXCEPT oe_clerk;
ORACLE
ENG
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Revoking Roles from Users
• Revoking roles from users requires the ADMIN OPTION or GRANT ANY ROLE privilege.
• To revoke a role:
REVOKE oe_clerk FROM scott;
REVOKE hr_manager FROM PUBLIC;
ORACLE
ENG
1:15 PM
12/2/2019


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Removing Roles
• Dropping a role:
– Removes it from all users and roles it was granted
– Removes it from the database
• Requires the ADMIN OPTION or DROP ANY ROLE privilege
• To drop a role:
DROP ROLE hr_manager;
ORACLE
ENG
1:16 PM
12/2/2019


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Obtaining Role Information
Information about roles can be obtained by querying the following views:
• DBA_ROLES: All roles that exist in the database
• DBA_ROLE_PRIVS: Roles granted to users and roles
• ROLE_ROL_PRIVS: Roles that are granted to roles
• DBA_SYS_PRIVS: System privileges granted to users and roles
• ROLE_SYS_PRIVS: System privileges granted to roles
• ROLE_TAB_PRIVS: Object privileges granted to roles
• SESSION_ROLES: Roles that the user currently has enabled
ORACLE
ENG
1:19 PM
12/2/2019

23) Creare rol fara parola:

SQL> create role emp_operation;
Role created.

24) Creare rol cu parola:

SQL> create role emp_operation_pas identified by student;

25) Asignare privilegii la rol:

SQL> grant insert, update on scott.emp_copy to emp_operation_pas;
Grant succeeded.

26) Asignare rol la un user:

SQL> grant emp_operation_pas to test;

27) Asignare rol la un user cu drept de reasignare catre alti useri:

SQL> grant emp_operation to test with admin option;

28) Asignare rol la toti userii:

SQL> grant emp_operation_pas to public;

29) Asignare rol la alt rol:
SQL> grant emp_operation to emp_operation_pas;
Grant succeeded.

30) Activare (enable) rol fara parola:

SQL> set role emp_operation;
Role set.

31) Activare (enable) rol cu parola:

SQL> set role emp_operation_pas

ERROR at line 1:
ORA-01979: missing or invalid password for rol 'EMP_OPERATION_PAS'

SQL> set role emp_operation_pas identified by student;
Role set.

32) Dezactivare parola pentru un rol:

SQL> alter role emp_operation_pas not identified;
Role altered.

33) Revocare rol pentru un user:

SQL>revoke emp_operation_pas from stud1;
Revoke succeeded.

34) Revocare rol pentru toti userii:

SQL>revoke emp_operation_pas from public;
Revoke succeeded.

35) Informatii din dictionar despre roluri create pe tabele:

SQL> desc role_tab_privs

Name	Null?	Type
ROLE	NOT NULL	VARCHAR2(30)
OWNER	NOT NULL	VARCHAR2(30)
TABLE_NAME	NOT NULL	VARCHAR2(30)
COLUMN_NAME		VARCHAR2(30)
PRIVILEGE	NOT NULL	VARCHAR2(40)
GRANTABLE		VARCHAR2(3)

SQL> select * from role_tab_privs where table_name='EMP';

ROLE	OWNER	TABLE_NAME	COLUMN_NAME	PRIVILEGE
EMP_OPERATION_PAS	SCOTT	EMP		SELECT
EMP_OPERATION_PAS	SCOTT	EMP		INSERT

36) Informatii din dictionar despre roluri active:

SQL> desc SESSION_ROLES

Name	Null?	Type
ROLE	NOT NULL	VARCHAR2(30)

SQL> select * from SESSION_ROLES;

```
ROLE
------------------------------
EMP_OPERATION_PAS
EMP_OPERATION
```

37) Informatii despre roluri parolate:

```
SQL> desc dba_roles

 Name
 -------------------
 ROLE
 PASSWORD_REQUIRED

SQL> select * from dba_roles;

ROLE                           PASSWORD
-------------------            --------
CONNECT                        NO
RESOURCE                       NO
DBA                            NO
SELECT_CATALOG_ROLE            NO
EXECUTE_CATALOG_ROLE           NO
DELETE_CATALOG_ROLE            NO
EXP_FULL_DATABASE              NO
IMP_FULL_DATABASE              NO
RECOVERY_CATALOG_OWN           NO
```

38) Informatii despre roluri asignate la alte roluri:

```
SQL> desc DBA_ROLE_PRIVS

 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
 GRANTEE                                            VARCHAR2(30)
 GRANTED_ROLE                                       NOT NULL VARCHAR2(30)
 ADMIN_OPTION                                       VARCHAR2(3)
 DEFAULT_ROLE                                       VARCHAR2(3)

SQL> select * from DBA_ROLE_PRIVS where grantee like 'EMP%';

GRANTEE                        GRANTED_ROLE                   ADM        DEF
------------------------------ ------------------------------ ---        ---
EMP_OPERATION_PAS              EMP_OPERATION                  NO         YES
```

39) Informatii despre privilegii de system acordate userilor :

```
SQL> desc DBA_SYS_PRIVS
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 GRANTEE                                   NOT NULL VARCHAR2(30)
 PRIVILEGE                                 NOT NULL VARCHAR2(40)
 ADMIN_OPTION                                       VARCHAR2(3)

SQL> select * from DBA_SYS_PRIVS where grantee='SCOTT';

GRANTEE                        PRIVILEGE                                ADM
------------------------------ ---------------------------------------- -------
SCOTT                          UNLIMITED TABLESPACE                     NO
```

40) Informatii despre privilegii pe obiecte acordate rolurilor:

```
SQL> desc  role_tab_privs

 Name                                      Null?    Type
 ----------------------------------------- -------- ------
 ROLE                                      NOT NULL VARCHAR2(30)
 OWNER                                     NOT NULL VARCHAR2(30)
 TABLE_NAME                                NOT NULL VARCHAR2(30)
 COLUMN_NAME                                        VARCHAR2(30)
 PRIVILEGE                                 NOT NULL VARCHAR2(40)
 GRANTABLE                                          VARCHAR2(3)

SQL> select role,owner,table_name, column_name, privilege from role_tab_privs
       where owner='SCOTT';

ROLE                           OWNER      TABLE_NAME     COLUMN_NAME     PRIVILEGE
------------------------------ ---------- -------------- --------------- ----------
EMP_OPERATION_PAS              SCOTT      EMP            EMPNO           UPDATE
EMP_OPERATION_PAS              SCOTT      EMP                            SELECT
EMP_OPERATION_PAS              SCOTT      EMP                            INSERT
EMP_OPERATION_PAS              SCOTT      EMP            ENAME           UPDATE
```

41) Informatii despre parametrii aferenti rolurilor:

```
SQL>  show parameters %roles

NAME                                 TYPE        VALUE
------------------------------------ ----------- -----------
max_enabled_roles                    integer     150
os_roles                             boolean     FALSE
remote_os_roles                      boolean     FALSE
```

42) Asignarea default a unui role catre un user:

SQL>alter user scott default role emp_operation_pas;

43) Asignarea default a tuturor rolurilor dar cu exceptii:

SQL>alter user scott default role ALL except emp_operation_pas;

44) Stergere rol din dictionar:

SQL> drop role emp_operation;